UNIT 3 FINAL PROJECT

Phishing Website Analysis

Exploratory Data Analysis and Dashboard of the PhiUSIIL Phishing URL Dataset

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1. Introduction

1.1 The dataset

Item	Details
Dataset title	PhiUSIIL Phishing URL Dataset
Source	UCI Machine Learning Repository: https://archive.ics.uci.edu/dataset/967/phiusiil+phishing+url+dataset
Creators	Arvind Prasad and Shalini Chandra (2024)
Size	235,795 websites (rows) and 55 columns: 54 features and 1 target
Target	`label`: 1 = legitimate (134,850), 0 = phishing (100,945)

The dataset describes real websites, and each one is labeled as phishing or legitimate. Some features come from the URL itself, such as its length and the number of digits and special characters. Others come from the webpage, such as whether it uses HTTPS, has a title, or has a password field. The dataset was loaded directly in Python with the `ucimlrepo` library.

1.2 Why I selected this dataset

- I worked on this project in October, which is Cybersecurity Awareness Month. Phishing is one of the most common cyber threats, so I wanted my analysis to help people understand how phishing websites look different from real ones.
- The dataset is large and recent, and it has a clear target, so it is easy to compare the two groups.
- It has different types of features (binary, numbers and text), which gives a lot to explore.

1.3 Problem definition

Phishing websites copy real websites to trick people into giving their passwords or bank details. The goal of this project is to find out what makes phishing websites different, and which signs can help people spot them. I explored seven questions:

#	Topic	Question
1	Class distribution	How many phishing and legitimate websites are there? Is the data balanced?
2	URL structure	Are phishing URLs longer? Do they have more digits, special characters or subdomains?
3	Security indicators	Do phishing websites use HTTPS? How often do they use an IP address instead of a domain name, or hide characters in the URL?
4	Page content	Are phishing pages smaller and simpler? Do they have a title, a favicon, a description or copyright information?
5	Sensitive elements	Do phishing pages have more password fields, hidden fields, or words like "bank", "pay" and "crypto"?
6	Top-level domains	Which endings (.com, .org, .app, ...) have the highest share of phishing websites?
7	Key features	Which features are most related to the label?

Before starting the analysis, I expected that:

- Phishing URLs will be **longer**, with more digits and special characters.
- Legitimate websites will almost **always use HTTPS**, while many phishing websites will not.
- Phishing pages will be **smaller and simpler**, because they are usually made quickly and removed fast.
- **Trust signals** like copyright info, social media links and a description will be much more common on legitimate websites.
- Some TLDs will have a much higher share of phishing than others.
- A few features will separate the two groups very clearly.

2. Data Exploration

In this step I loaded the data and checked its structure with pandas: the first rows (`df.head()`), the shape (`df.shape`), the data types (`df.info()`), the statistics (`df.describe()`), the missing values (`df.isnull().sum()`) and the duplicates.

2.1 Structure

- **Shape:** 235,795 rows and 55 columns (54 features and the target).
- **Data types:** 41 integer columns, 10 decimal (float) columns and 4 text columns (`URL`, `Domain`, `TLD`, `Title`).
- **Binary features:** 19 features only contain 0 and 1, such as `IsHTTPS` and `HasPasswordField`. Two of them, `NoOfURLRedirect` and `NoOfSelfRedirect`, are named like counts but only contain 0 and 1.
- **Target:** `label` is 57.2% legitimate and 42.8% phishing, so the classes are reasonably balanced.

The first rows of the dataset (a few columns only):

URL	URLLength	TLD	IsHTTPS	LineOfCode	HasTitle	label
https://www.southbankmosaics.com	31	com	1	558	1	1
https://www.uni-mainz.de	23	de	1	618	1	1
https://www.voicefmradio.co.uk	29	uk	1	467	1	1
https://www.sfnmjournal.com	26	com	1	6,356	1	1
https://www.rewildingargentina.org	33	org	1	6,089	1	1

2.2 Data dictionary

I grouped the 55 columns by what they describe. The same groups were used to organize the EDA. Columns marked with * were renamed in Section 3.4.

Column	Type	Description
1. Identifiers (3 columns)		
`URL`	Text	The full web address of the website
`Domain`	Text	The domain part of the URL (for example, www.google.com)
`TLD`	Text	The last part of the domain (for example, com, org, uk)
2. URL structure (13 columns)		
`URLLength`	Count	Number of characters in the URL
`DomainLength`	Count	Number of characters in the domain
`TLDLength`	Count	Number of characters in the TLD
`NoOfSubDomain`	Count	Number of subdomains (for example, www in www.google.com)

Column	Type	Description
`NoOfLettersInURL`	Count	Number of letters in the URL
`LetterRatioInURL`	Ratio	Letters divided by URL length
`NoOfDegitsInURL *`	Count	Number of digits in the URL
`DegitRatioInURL *`	Ratio	Digits divided by URL length
`NoOfEqualsInURL`	Count	Number of = signs in the URL
`NoOfQMarkInURL`	Count	Number of ? signs in the URL
`NoOfAmpersandInURL`	Count	Number of & signs in the URL
`NoOfOtherSpecialCharsInURL`	Count	Number of other special characters in the URL
`SpacialCharRatioInURL *`	Ratio	Special characters divided by URL length
3. Security indicators (5 columns)		
`IsDomainIP`	Binary	1 if the URL uses an IP address instead of a domain name
`IsHTTPS`	Binary	1 if the website uses HTTPS (corrected in Section 3.4)
`HasObfuscation`	Binary	1 if the URL contains hidden (encoded) characters, such as %20
`NoOfObfuscatedChar`	Count	Number of encoded characters in the URL
`ObfuscationRatio`	Ratio	Encoded characters divided by URL length
4. Derived scores (6 columns)		
`URLSimilarityIndex`	Score (0–100)	How similar the URL is to a list of known legitimate URLs
`CharContinuationRate`	Ratio	How much of the URL is made of long runs of the same character type (letters, digits or symbols)
`TLDLegitimateProb`	Probability	How often the TLD appears among legitimate websites
`URLCharProb`	Probability	How common the URL's characters are in legitimate URLs
`DomainTitleMatchScore`	Score (0–100)	How similar the page title is to the domain
`URLTitleMatchScore`	Score (0–100)	How similar the page title is to the URL
5. Page size and structure (12 columns)		
`LineOfCode *`	Count	Number of lines in the page's source code
`LargestLineLength`	Count	Number of characters in the longest line of code
`NoOfImage`	Count	Number of images on the page
`NoOfCSS`	Count	Number of CSS (style) files
`NoOfJS`	Count	Number of JavaScript files
`NoOfSelfRef`	Count	Number of links to the same website
`NoOfEmptyRef`	Count	Number of empty links (links that go nowhere)
`NoOfExternalRef`	Count	Number of links to other websites
`NoOfiFrame`	Count	Number of iframes (other pages embedded inside the page)

Column	Type	Description
`NoOfPopup`	Count	Number of pop-up windows
`NoOfURLRedirect *`	Binary	1 if the page redirects to another URL
`NoOfSelfRedirect *`	Binary	1 if the page redirects to another page on the same website
6. Trust signals (8 columns)		
`HasTitle`	Binary	1 if the page has a title
`Title`	Text	The page title (cleaned in Section 3.1)
`HasFavicon`	Binary	1 if the page has a favicon (the small icon in the browser tab)
`HasDescription`	Binary	1 if the page has a description for search engines
`HasCopyrightInfo`	Binary	1 if the page shows copyright information
`HasSocialNet`	Binary	1 if the page links to social media
`IsResponsive`	Binary	1 if the page adapts to mobile screens
`Robots *`	Binary	1 if the website has a robots.txt file (instructions for search engines)
7. Sensitive elements (7 columns)		
`HasPasswordField`	Binary	1 if the page has a password field
`HasHiddenFields`	Binary	1 if the page has hidden form fields
`HasSubmitButton`	Binary	1 if the page has a submit button
`HasExternalFormSubmit`	Binary	1 if a form on the page sends data to another website
`Bank *`	Binary	1 if the word "bank" appears on the page
`Pay *`	Binary	1 if the word "pay" appears on the page
`Crypto *`	Binary	1 if the word "crypto" appears on the page
8. Target (1 columns)		
`label`	Binary	1 = legitimate, 0 = phishing

2.3 Summary statistics

Several features are heavily right-skewed, with a few extreme values far above the rest:

Feature	Median	Maximum
URLLength (characters)	27	6,097
LineOfCode	429	442,666
LargestLineLength (characters)	1,090	13,975,732

These statistics are computed on the raw data (235,795 rows). The values in Section 4 and in the dashboard are computed after cleaning, so small differences are expected (for example, the median lines of code is 429 before cleaning and 431 after).

Because the features are heavily skewed, most charts in the EDA use a log scale or a limited axis range.

2.4 Data quality issues found

`df.isnull().sum()` reported **no missing values**, but a closer look at the text columns and duplicates showed several hidden problems:

- **Hidden missing values in `Title`:** 32,719 pages store `"0"` instead of a title, and 21 titles appear as `"#NAME?"`, an Excel formula error. Comparing with `HasTitle` confirmed that `"0"` means "no title" (32,714 of the 32,719 rows have `HasTitle = 0`).
- **Values in `TLD` that are not real TLDs:** 599 values are numbers (for URLs that use an IP address, the last number of the IP was stored as the TLD) and 26 values include a port number (for example `com:443`). 41 values are Punycode TLDs (`xn--`), which are real TLDs written in non-Latin characters.
- **Duplicated URLs:** there are no fully duplicated rows, but 425 URLs appear twice. All 850 rows are phishing, and the two copies differ slightly in some features.

3. Data Cleaning

All cleaning steps were applied to a copy of the original data. For each problem I considered several options and chose the one that keeps the most information.

3.1 Hidden missing values

Rows without a title were **not deleted**, because a missing title could itself be a sign of phishing. Instead, `"0"` was replaced with `"No Title"` (32,714 rows), and the 5 inconsistent rows (`HasTitle = 1` but `Title = "0"`) and the 21 Excel errors were replaced with `"Unknown Title"`, because a title exists but its value is unknown.

3.2 URL duplicates

Before deleting anything, I checked whether each pair is really the same observation:

- In **74 pairs**, the page itself changed between the two collections (for example, an almost empty page in one copy and a full login page in the other). These are two different observations, so both were kept.
- In the other **351 pairs**, the page content is identical, and `URLLength` differs by exactly one character. One copy's URL features were calculated without the last character of the URL. This also happens in 79% of all rows, so it comes from how the features were extracted.
- I then checked whether the shorter version could be a real, different address. For 286 pairs it could not (for example, `site.com/` and `site.com` are the same address), so these **286 true duplicates were removed**. The other 65 pairs were kept.

Removing the duplicates changed the phishing averages by less than 1%.

3.3 Outliers

70.5% of rows have at least one outlier by the IQR rule, so removing them would delete most of the dataset. No impossible values were found (no negative values, no ratios outside 0–1 and no scores outside 0–100), so the outliers are real values, not errors. **All outliers were kept**, and Section 4.2 shows that they are strongly linked to the label.

3.4 Other fixes

- **Renaming:** 10 columns were renamed to fix spelling mistakes (`Degit` → `Digit`, `Spacial` → `Special`) and to make names clearer (`LineOfCode` → `NoOfLinesOfCode`, `Bank` → `HasBankKeyword`, `NoOfURLRedirect` → `HasURLRedirect`, …).
- **TLD values:** port numbers were removed (`com:443` → `com`), and the 607 IP numbers (the 599 found in Section 2.4, plus 8 IP addresses with a port such as `240:8087` that became visible after the port was removed) were replaced with `"IP"`. A new `TLDType` column groups all TLDs into 6 types: classic generic, country code, new generic, restricted, IP address and non-Latin (Punycode).
- **Fixing `IsHTTPS`:** 493 URLs that start with `http://` had `IsHTTPS = 1`, because the word "https" appeared somewhere else in the URL (for example, `http://www.httpsbinance.com`). All 493 are phishing. The column was corrected so it is 1 only when the URL starts with `https://`.

- **Readable label:** a `label_name` column ("Phishing" / "Legitimate") was added for charts and the dashboard.
- **Data types:** all columns already had suitable types, so no conversion was needed.

3.5 Cleaning summary

Problem	Finding	Action
Hidden missing values	Title stores missing titles as "0" and has 21 "#NAME?" errors	"No Title" (32,714) and "Unknown Title" (26)
Duplicates	425 URLs appear twice, all phishing	286 true duplicates removed; 139 pairs kept
Outliers	70.5% of rows have an IQR outlier; no impossible values	Kept (real values)
Column names	Spelling mistakes and unclear names	10 columns renamed
TLD values	26 include a port number; 607 are IP numbers	Ports removed; IPs replaced with "IP"; TLDType added
HTTPS flag	493 http:// URLs marked as HTTPS	Corrected from the URL
Label	0/1 values are easy to mix up	label_name added

Final dataset: 235,509 rows and 57 columns (100,659 phishing and 134,850 legitimate), saved as `phishing_clean.csv`.

4. Exploratory Data Analysis

All charts use the same colors: **orange for phishing** and **blue for legitimate** websites. The section starts with univariate analysis (one feature at a time), then bivariate analysis (each feature compared by label), and ends with multivariate analysis.

4.1 Univariate analysis

Class distribution (Question 1)

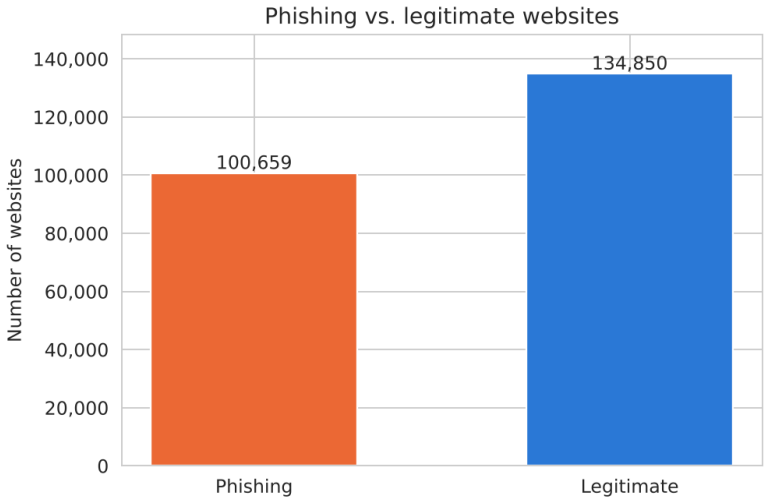


Figure 1. *Number of phishing and legitimate websites after cleaning. The dataset has 100,659 phishing (42.7%) and 134,850 legitimate (57.3%) websites, so the classes are fairly balanced.*

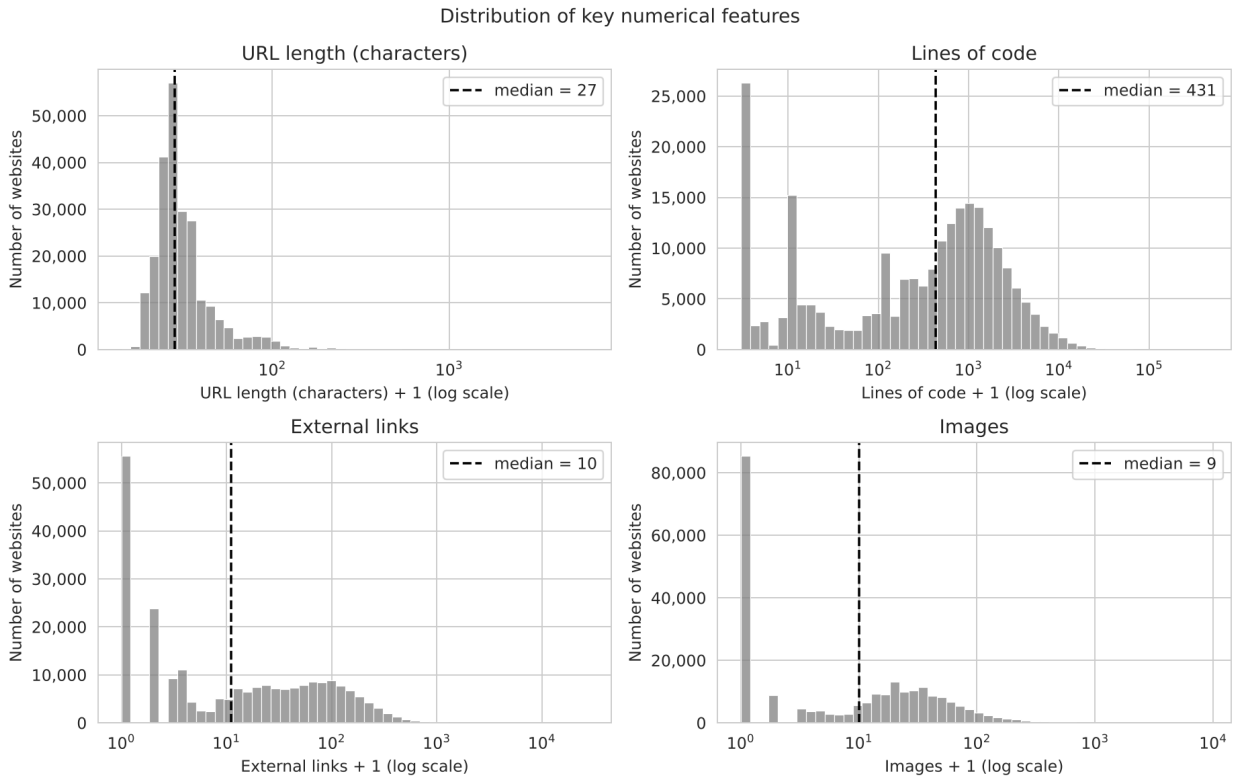


Figure 2. *Distribution of four key numerical features (log scale, 1 added to every value so that zeros can be shown). All four are right-skewed. Lines of code has two peaks: very small pages (26% have 20 lines or fewer) and pages around 1,000 lines. External links and images are 0 for 24% and 36% of websites.*

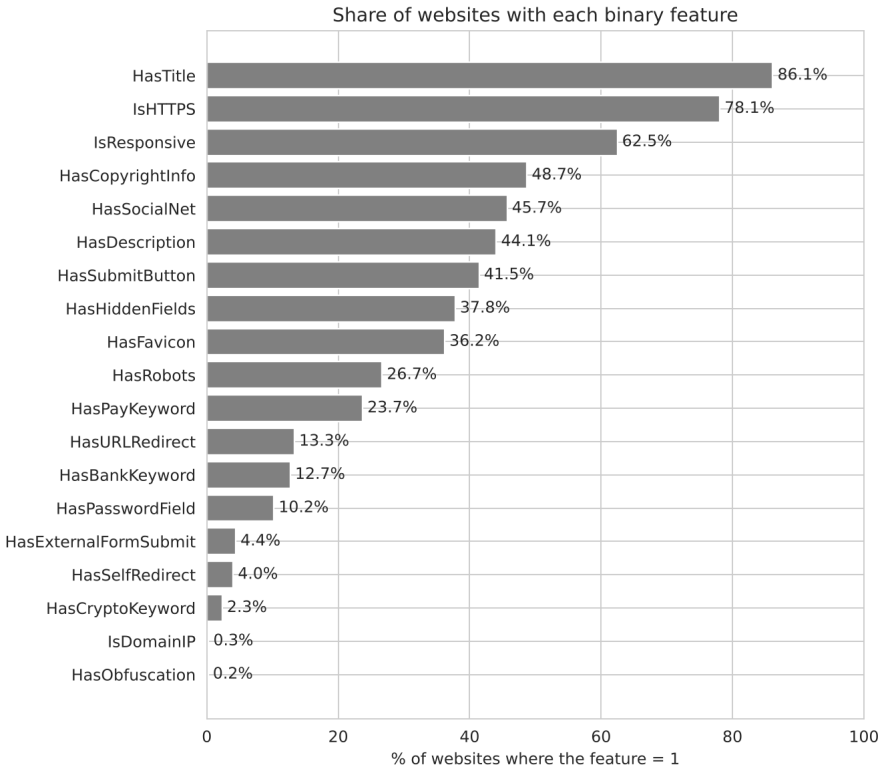


Figure 3. *Share of websites where each binary feature is 1. 86.1% have a title and 78.1% use HTTPS, while IP addresses (0.3%) and obfuscated characters (0.2%) are rare.*

4.2 Bivariate analysis

URL structure (Question 2)

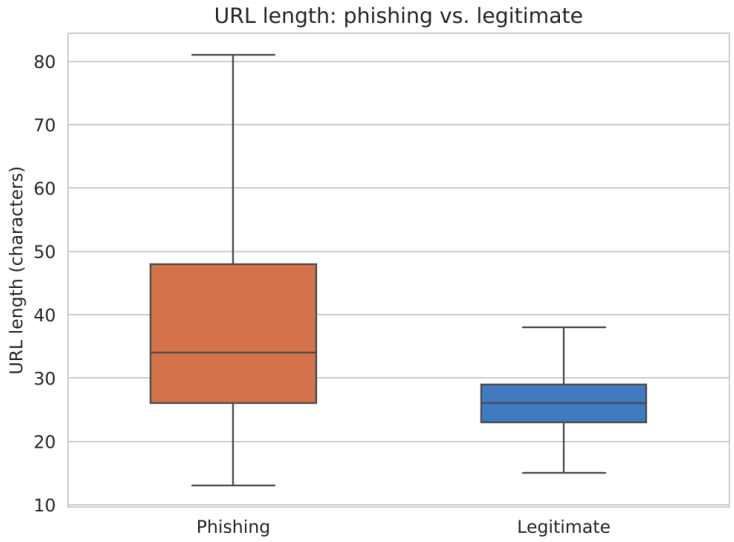


Figure 4. *URL length by class (outliers hidden). Phishing URLs are longer (median 34 vs 26 characters) and much more spread out: 22% are longer than 50 characters, compared to none of the legitimate URLs.*

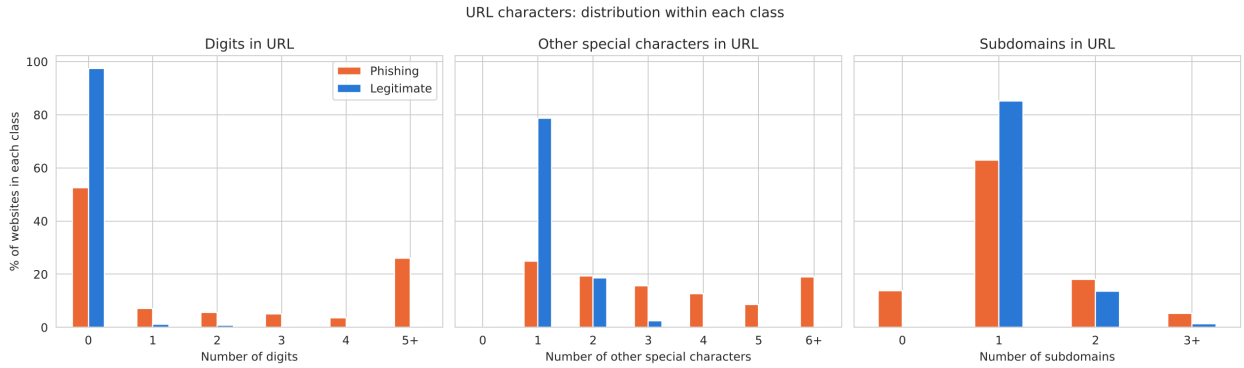


Figure 5. *Digits, special characters and subdomains in the URL, within each class. 97% of legitimate URLs have no digits, compared to 53% of phishing URLs, and 26% of phishing URLs have 5 or more. 14% of phishing URLs have no subdomain, while every legitimate URL has at least one (www).*

Limitation: every legitimate URL in this dataset is a homepage in the form https://www.domain, while 60% of phishing URLs include a path. Part of these differences comes from how the legitimate URLs were collected.

Security indicators (Question 3)

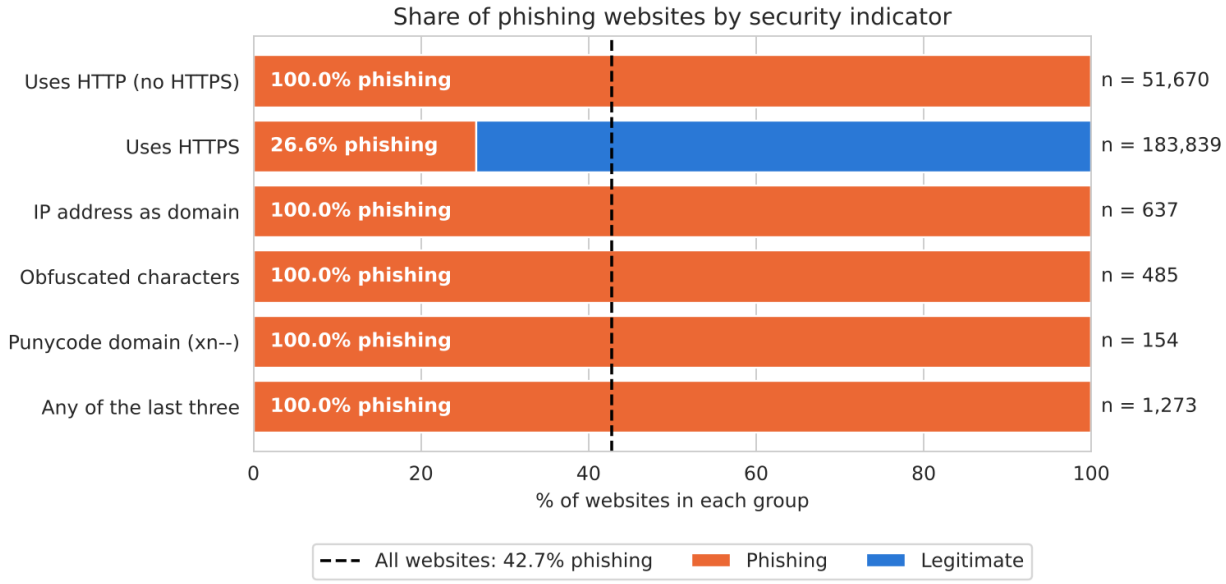


Figure 6. *Share of phishing websites for each security indicator. Every website without HTTPS (51,670) is phishing, but HTTPS is not a guarantee of safety: 26.6% of HTTPS websites are still phishing, and 49% of all phishing websites use HTTPS. IP addresses, obfuscated characters and Punycode domains are 100% phishing, but together they cover only 1.3% of phishing websites.*

Page content and trust signals (Question 4)

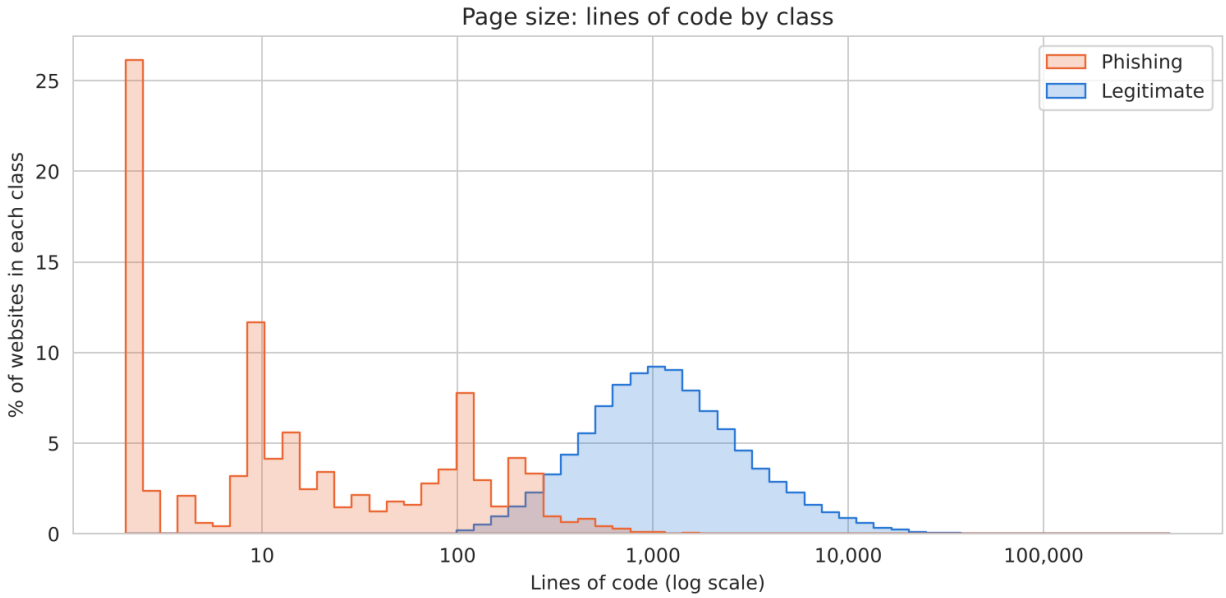


Figure 7. *Lines of code by class. The two peaks from Figure 2 are explained by the label: phishing pages have a median of 12 lines of code, compared to 1,105 for legitimate pages, and 62% of phishing pages have 20 lines or fewer. No legitimate page has fewer than 102 lines.*

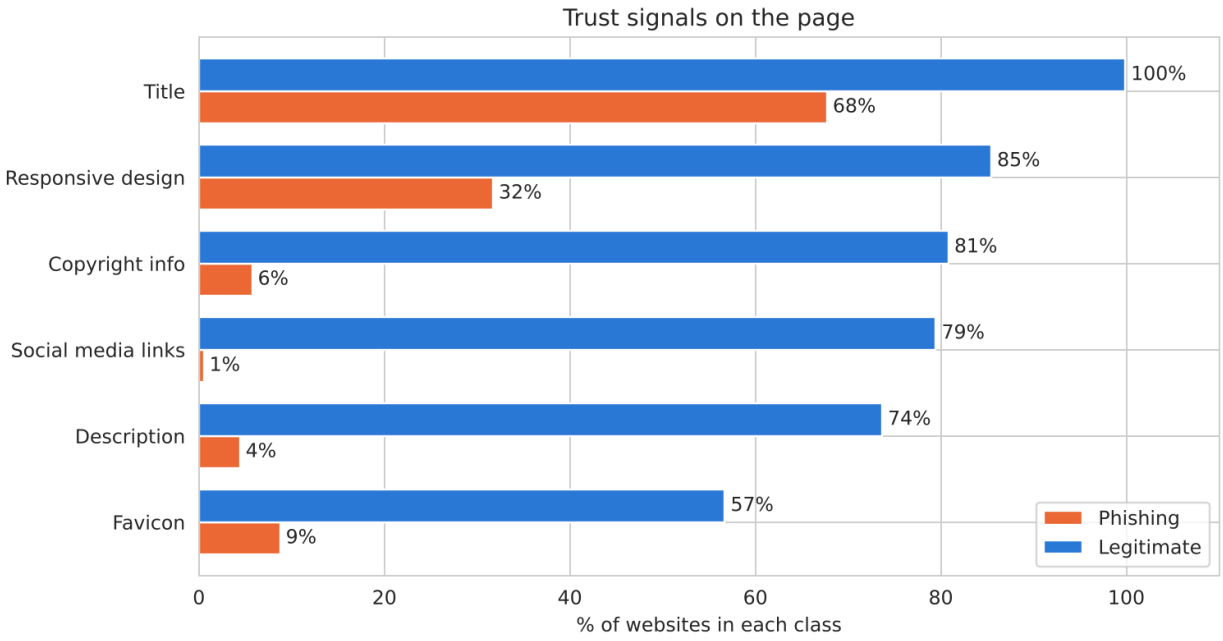


Figure 8. *Trust signals by class. The largest gap is in social media links (under 1% of phishing pages vs 79% of legitimate pages), followed by copyright info (6% vs 81%) and a description (4% vs 74%). 99.5% of pages without a title are phishing.*

Sensitive elements (Question 5)

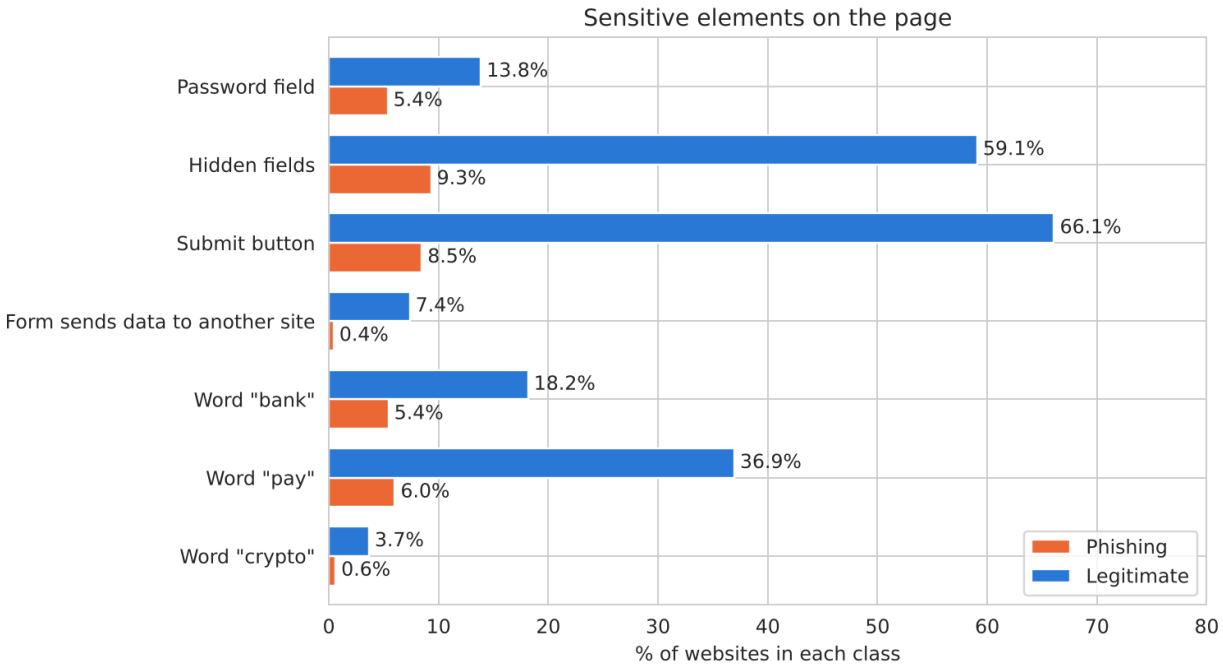


Figure 9. *Sensitive elements by class. Contrary to the expectation, every element is more common on legitimate websites, for example password fields (13.8% vs 5.4%).*

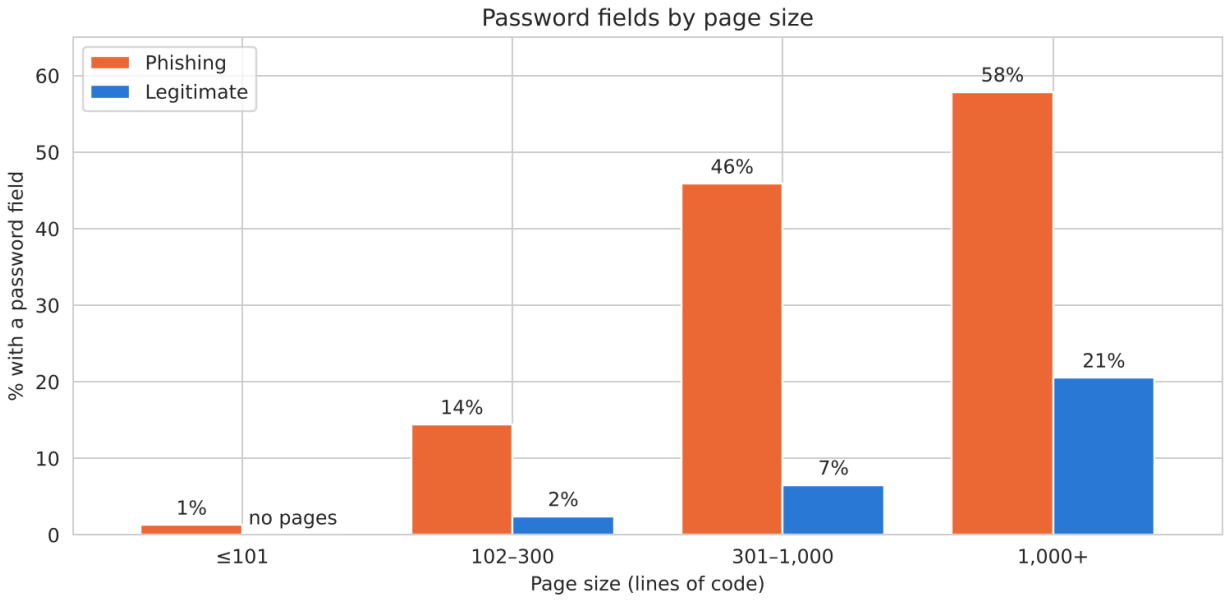


Figure 10. *Password fields by page size. When pages of similar size are compared, the pattern reverses: phishing pages have password fields far more often (for example, 46% vs 7% for pages with 301–1,000 lines). The overall result was misleading because most phishing pages are too small to contain a form. Only 280 phishing pages have 1,000+ lines of code, so that group is small.*

Top-level domains (Question 6)

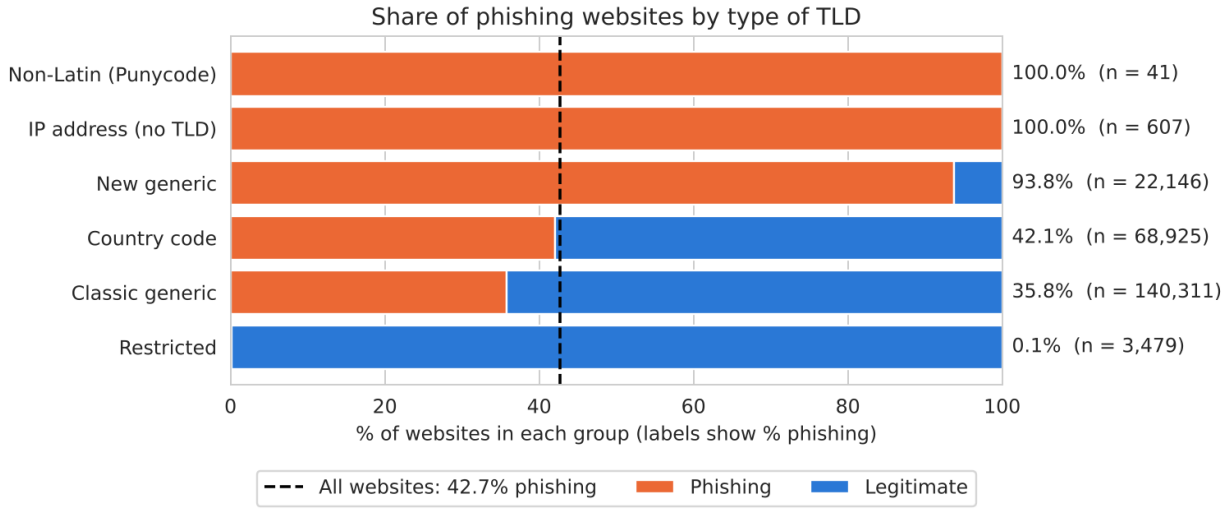


Figure 11. *Share of phishing websites by type of TLD. New generic TLDs (.app, .dev, .top, ...) are 93.8% phishing, more than double the average, while restricted TLDs (.gov, .edu, .mil, .int) are almost never phishing (0.1%).*

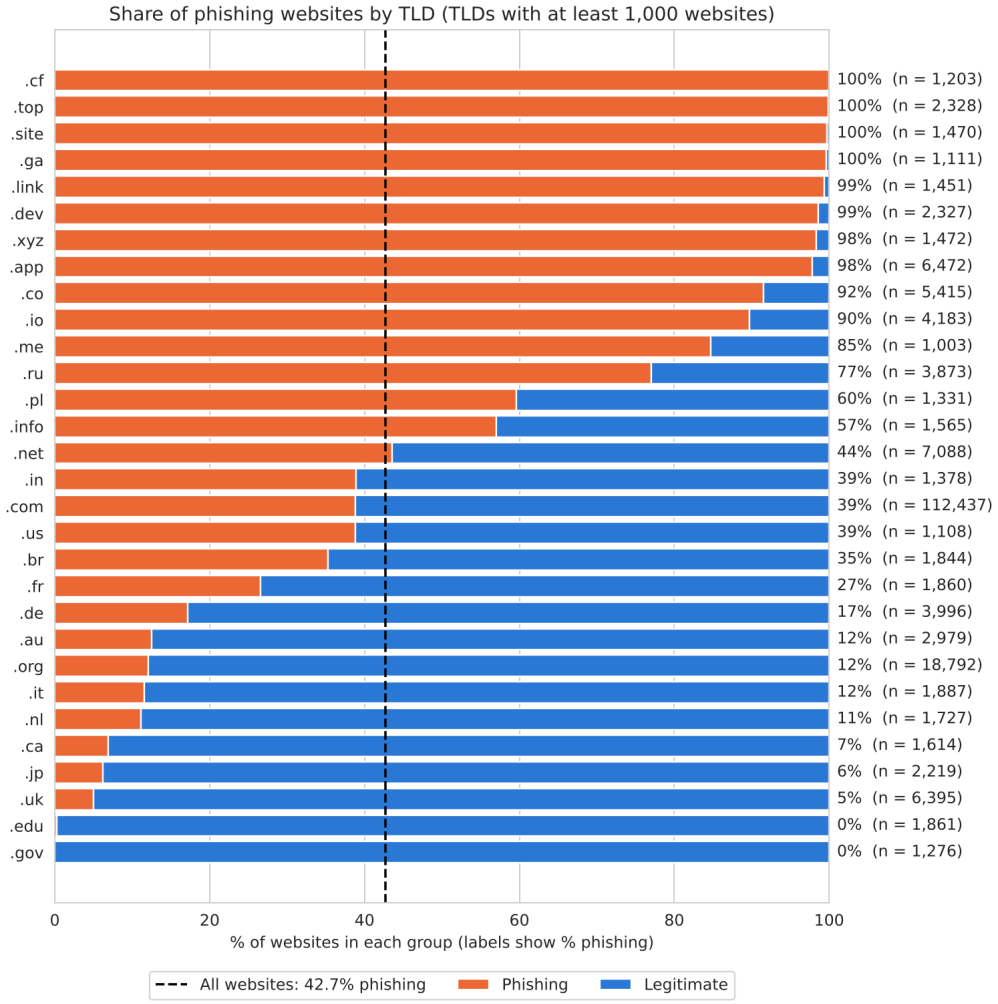


Figure 12. *Share of phishing websites for TLDs with at least 1,000 websites. Eight TLDs are 98–100% phishing. Many phishing websites on these TLDs use free hosting platforms: Firebase (web.app, 90% of phishing on .app), Replit (repl.co, 76%), Cloudflare (workers.dev, 62%) and IPFS (ipfs.io, 48%). .com holds almost half of all websites, and its phishing share (39%) is close to the average.*

Outliers by label

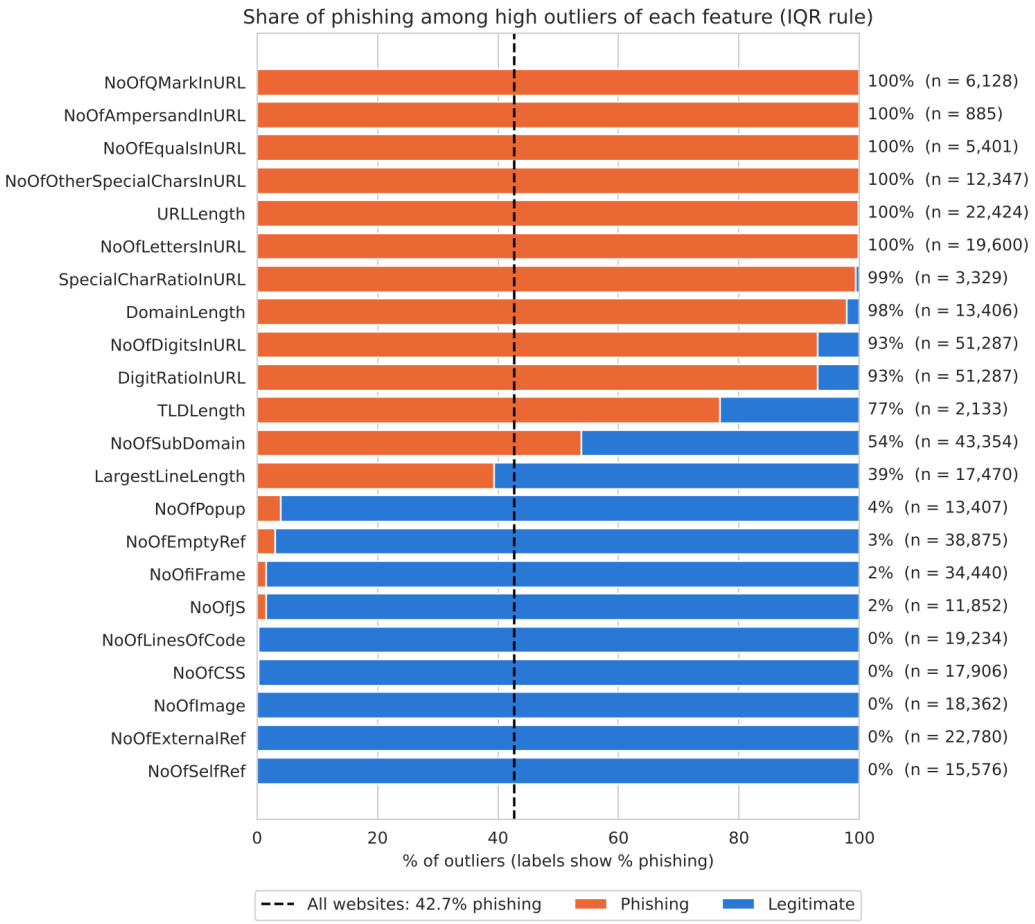


Figure 13. *Share of phishing among the high outliers of each feature. URL-based outliers (long URLs, many digits or special characters) are 93–100% phishing, while page-content outliers (many lines of code, images or links) are 96–100% legitimate. The outliers are not noise, but some of the clearest differences between the two classes.*

4.3 Multivariate analysis

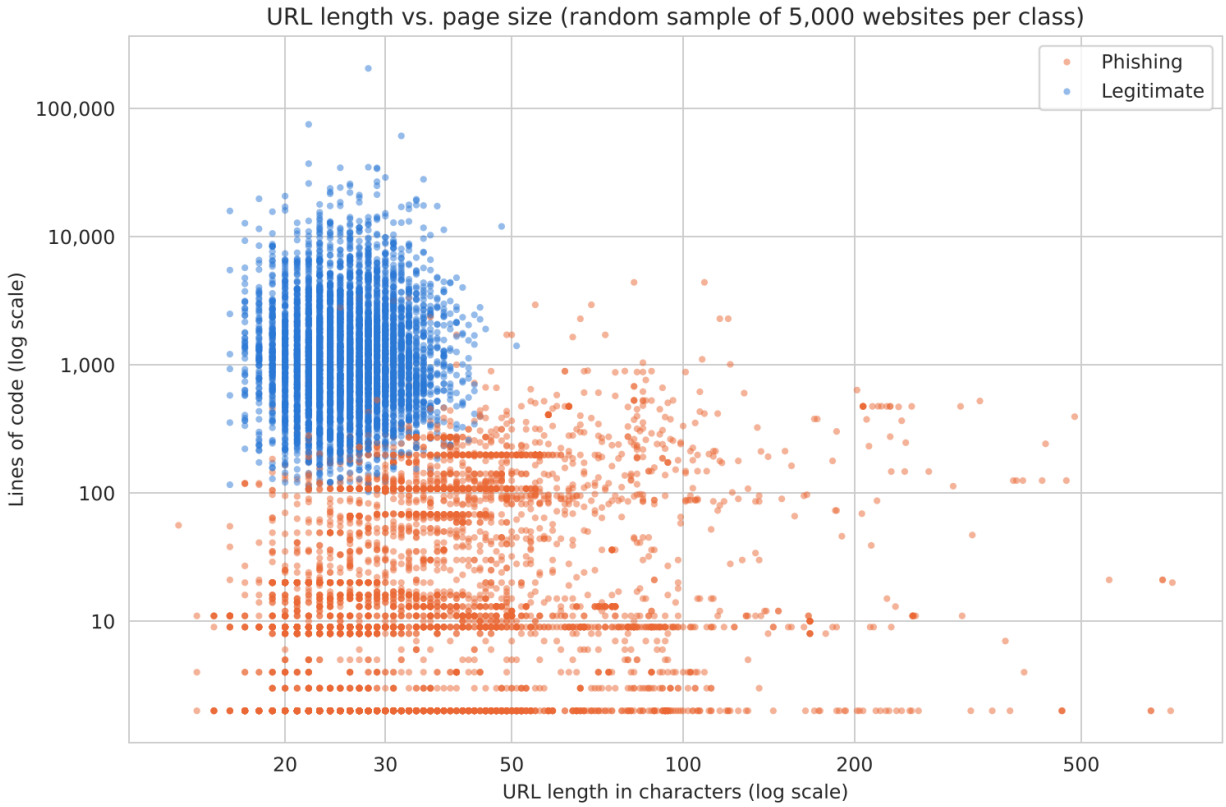


Figure 14. *URL length vs. page size (random sample of 5,000 websites per class). Legitimate websites form one tight cluster of short URLs and large pages, while phishing websites are spread out. The horizontal lines are many phishing pages with exactly the same size (26,315 have exactly 2 lines of code). 23.2% of phishing pages are as large as legitimate pages, so page size alone is not enough.*

Key features (Question 7)

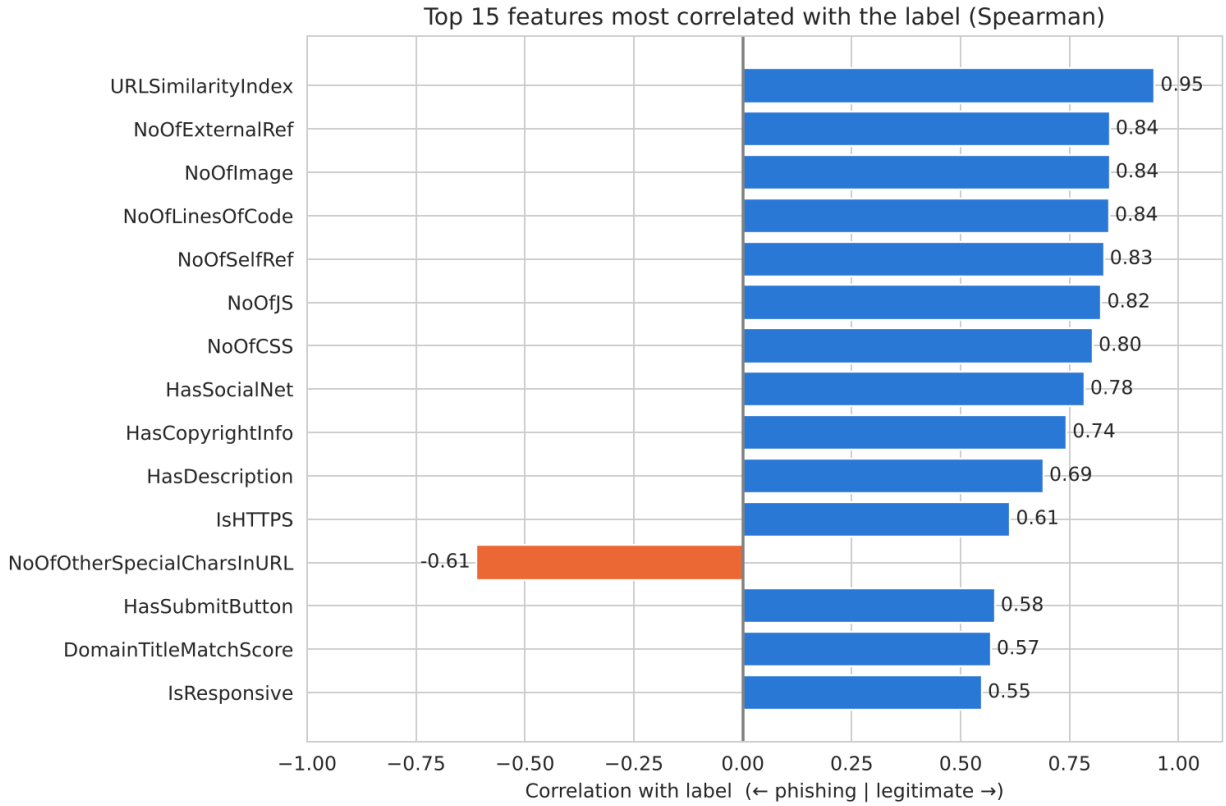


Figure 15. *The 15 features most correlated with the label (Spearman, used because the features are skewed). URLSimilarityIndex is the strongest (0.95), followed by page-size features (0.80–0.84) and trust signals. NoOfOtherSpecialCharsInURL (−0.61) is the strongest sign of phishing.*

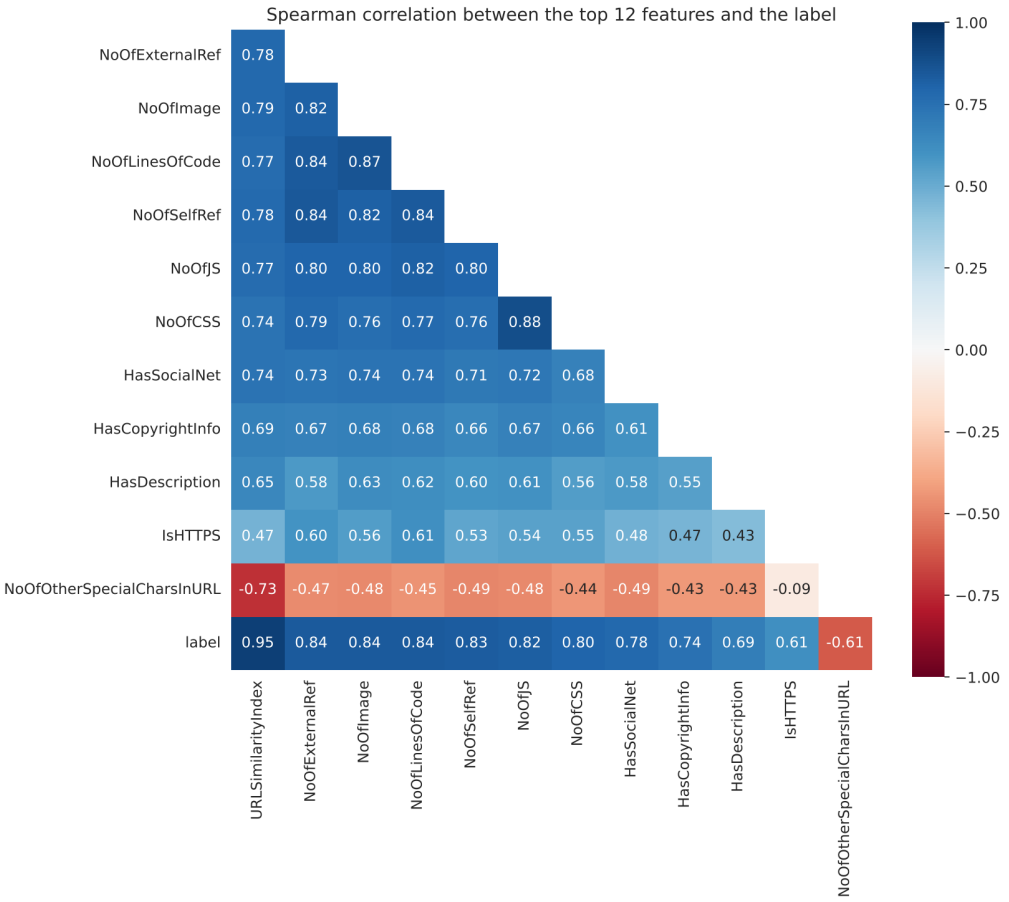


Figure 16. *Spearman correlation between the top 12 features and the label. The page-size features are strongly correlated with each other (0.76–0.88), so they mostly carry the same information.*

Note: every legitimate website has a URLSimilarityIndex of exactly 100, which suggests the legitimate websites come from the same reference list used to compute this score. This makes the feature very strong, but partly circular.

5. Dashboard Overview

I built an interactive dashboard with Streamlit and Plotly to explore the cleaned data. It is run with `streamlit run app/app.py`. The app reads `phishing_clean.csv.gz`. The cleaned dataset is stored as `phishing_clean.csv.gz`, a gzip-compressed copy of `phishing_clean.csv`, so that it stays under GitHub's file size limit.

Sidebar: a short description of the dataset and three filters: a dropdown for the class, a multiselect for the TLD type and a slider for the URL length. Every filter updates all the numbers, tables and charts on the page.

Design: the dashboard uses a dark navy and cyan theme inspired by security operations centers (SOC), with a matching light version. The charts keep the same colors as the rest of the report: orange for phishing and blue for legitimate websites.

Main page: four key numbers (websites, % phishing, % HTTPS and median URL length) and three tabs:

- **Data:** a preview of the filtered data and summary statistics by class.
- **Charts:** interactive charts. The user chooses any feature and compares it between the two classes, plus the phishing share by TLD type and the security indicators.
- **Insights:** the main findings of the analysis and the limitations of the data.

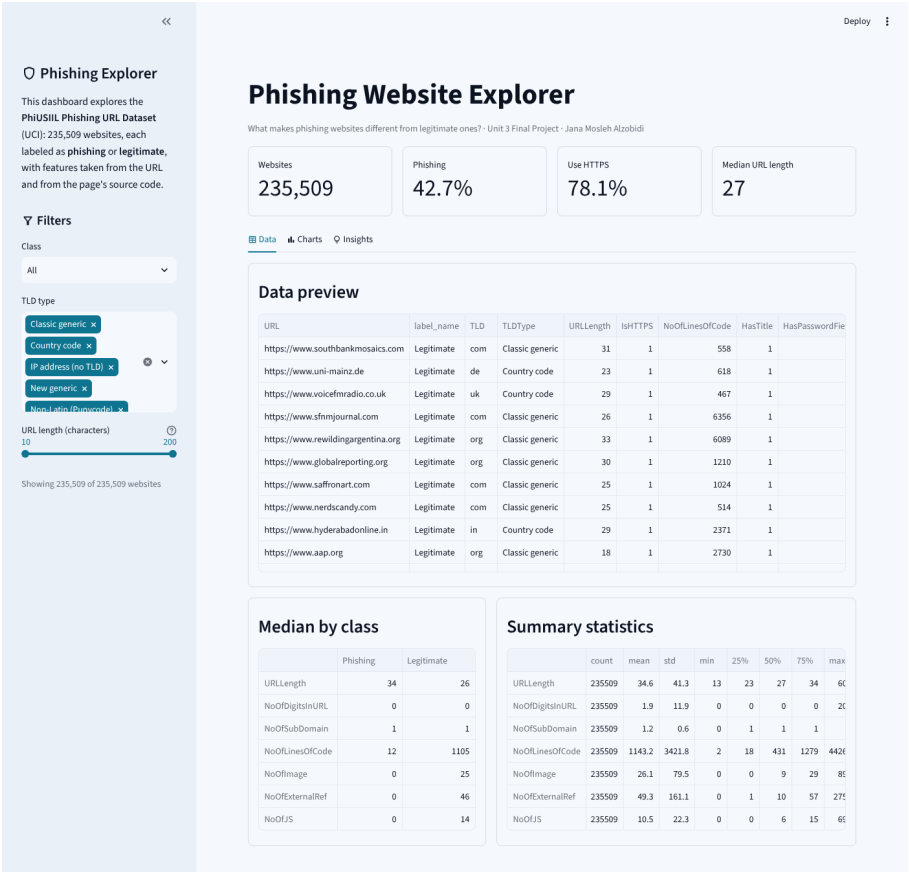


Figure 17. *Dashboard: sidebar with the filters, the key numbers and the Data tab.*



Figure 18. *Dashboard: the Charts tab (main area). The selected feature (URLLength) is compared between the two classes.*

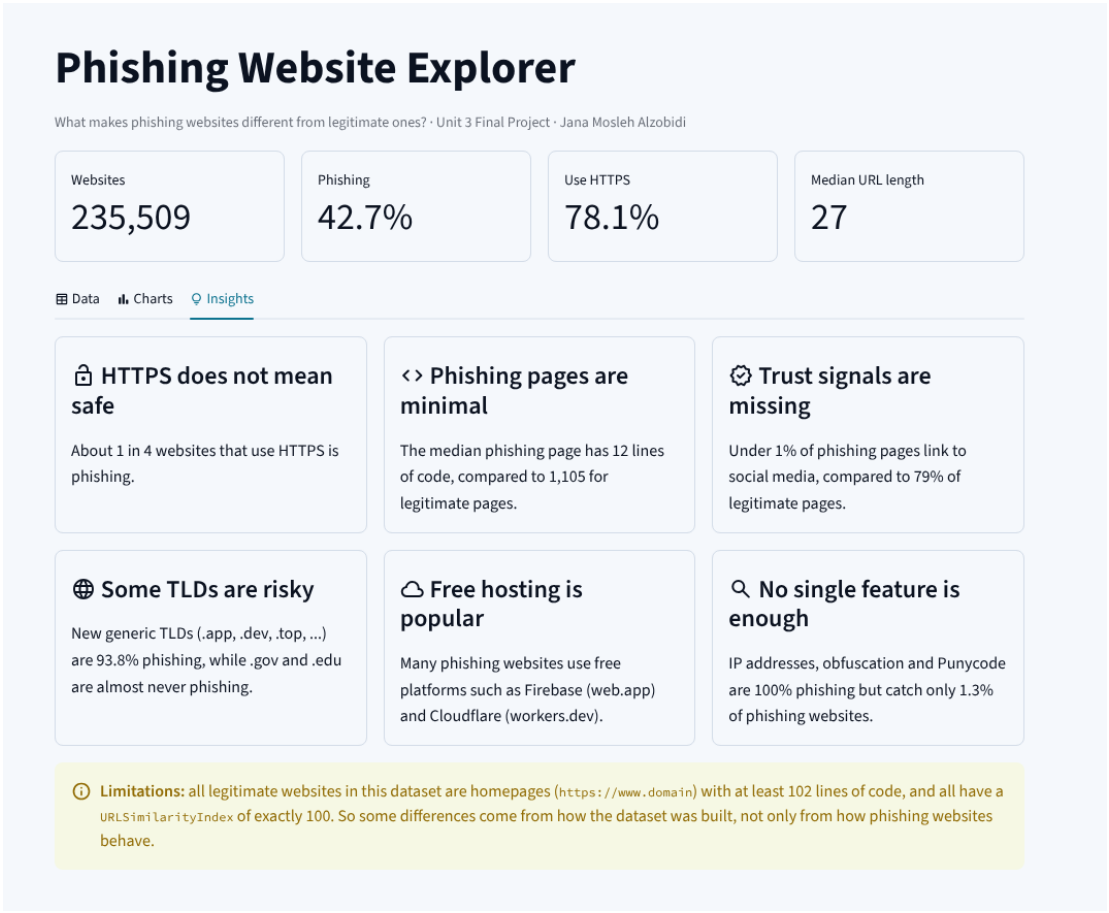


Figure 19. *Dashboard: the Insights tab (main area). Each finding is shown as a card, and the limitations are shown below.*



Figure 20. *Dashboard in dark mode. The theme follows the viewer's system setting and can also be changed from the app menu (Settings, then Theme).*

6. Insights and Conclusion

6.1 Answers to the questions

The table answers each question from Section 1.3, next to what I expected before the analysis, and shows whether my expectation was confirmed.

Question	My expectation	Answer	Result
Q1. Class distribution: How many phishing and legitimate websites are there? Is the data balanced?	No expectation	100,659 phishing (42.7%) and 134,850 legitimate (57.3%) websites. The classes are fairly balanced.	–
Q2. URL structure: Are phishing URLs longer? Do they have more digits, special characters or subdomains?	Phishing URLs will be longer, with more digits and special characters.	Yes. Median length 34 vs 26 characters; 47% of phishing URLs contain digits vs 3% of legitimate ones; 19% of phishing URLs have 6 or more special characters. Subdomains are mixed: 14% of phishing URLs have none, while every legitimate URL has at least one (www).	**Confirmed**
Q3. Security indicators: Do phishing websites use HTTPS? How often do they use an IP address, or hide characters in the URL?	Legitimate websites will almost always use HTTPS, while many phishing websites will not.	49% of phishing websites use HTTPS, compared to 100% of legitimate ones, but 1 in 4 HTTPS websites is still phishing. IP addresses, obfuscated characters and Punycode are 100% phishing, but cover only 1.3% of phishing websites.	**Confirmed**
Q4. Page content: Are phishing pages smaller and simpler? Do they have a title, a favicon, a description or copyright information?	Phishing pages will be smaller and simpler, and trust signals will be much more common on legitimate websites.	Yes. Median of 12 lines of code vs 1,105. Social media links on under 1% of phishing pages vs 79% of legitimate pages; 99.5% of pages without a title are phishing.	**Confirmed**
Q5. Sensitive elements: Do phishing pages have more password fields, hidden fields, or words like "bank", "pay" and "crypto"?	These elements will be more common on phishing pages.	Overall, no: all of them are more common on legitimate pages (password fields 13.8% vs 5.4%). But among pages of similar size, phishing pages have password fields 3 to 7 times more often.	**Partly**
Q6. Top-level domains: Which endings have the highest share of phishing websites?	Some TLDs will have a much higher share of phishing than others.	New generic TLDs are 93.8% phishing, and 8 TLDs (.cf, .top, .site, .ga, .link, .dev, .xyz, .app) are 98–100% phishing. .gov and .edu are almost 0%.	**Confirmed**
Q7. Key features: Which features are most related to the label?	A few features will separate the two groups very clearly.	URLSimilarityIndex (0.95), the page-size features (0.80–0.84) and the trust signals. Some of them are partly affected by how the data was collected.	**Confirmed**

6.2 Key takeaways

- **HTTPS does not mean safe.** The padlock only means the connection is encrypted, and about 1 in 4 HTTPS websites in this dataset is phishing.
- **Phishing pages are minimal.** 62% have 20 lines of code or fewer, and long, complex URLs point to phishing, while large, content-rich pages point to legitimate websites.

- **Many phishing websites use free and cheap services**, such as free hosting platforms (Firebase, Cloudflare), which are quick to set up and come with HTTPS, and cheap TLDs such as .top and .xyz.
- **Page size hides the real pattern for sensitive elements:** among pages of similar size, phishing pages ask for a password 3 to 7 times more often.
- **No single feature is enough.** The rarest signs (IP addresses, obfuscation, Punycode) are 100% phishing but catch only 1.3% of phishing websites.
- **Simple checks go a long way:** an unusual TLD, a missing title, no social media or copyright information, and a long URL with many special characters are all strong warning signs.

6.3 What I learned

This project taught me many new things about phishing, about how URLs work, and about EDA methods.

About phishing

- **HTTPS does not mean a website is safe.** The padlock only means the connection is encrypted. Phishers can get HTTPS for free, and almost half of the phishing websites in this dataset use it.
- **Phishers use free services** such as Firebase, Cloudflare Workers and Replit, because they are quick to set up, cost nothing and give the website a trusted-looking address with HTTPS.
- **Phishers hide or fake the meaning of a URL**, for example by putting the word "https" inside the domain name (http://www.httpsbinance.com) to make a site look secure.
- **Homograph attacks:** a type of phishing where letters in a domain are replaced with letters from another alphabet that look the same, for example a Russian "a" instead of the English "a" in "apple.com". The fake domain looks legitimate to the user, but it is a completely different address.

About URLs

- **The parts of a URL:** the protocol (http:// or https://), the subdomain (www), the domain, the top-level domain (TLD) and the path. I learned that the protocol is only set by how the URL starts, which is how I found the IsHTTPS error.
- **Ports:** a URL can include a port number after the domain (site.com:443), which tells the browser which "door" of the server to use.
- **Types of TLDs:** classic generic (.com, .org), country codes (.uk, .sa), new generic (.app, .top) and restricted TLDs (.gov, .edu) that only verified organizations can register.
- **Punycode:** domain names with non-Latin letters (such as Arabic or Russian) are stored in a special format that starts with `xn--`, followed by the foreign letters written in Latin characters. For example, the Russian TLD .pф is stored as `xn--p1ai`. This is also how homograph domains appear in the data.
- **Encoded characters (obfuscation):** special characters in a URL can be written as % followed by a code, for example %20 for a space, %2F for / and %40 for @. Phishers use this to hide what the URL really contains.

About EDA

- **Hidden missing values:** a dataset with no missing values can still have placeholders, such as "0" and "#NAME?" in the Title column.

- **Simpson's paradox:** a pattern can reverse when the data is split into groups. Overall, password fields were more common on legitimate pages, but among pages of similar size they were much more common on phishing pages.
- **Spearman vs. Pearson correlation:** Pearson only measures linear relationships and is affected by extreme values, while Spearman compares ranks, so it works better for skewed features (0.84 vs 0.27 for lines of code).
- **Log scales** make skewed features readable, and adding 1 before plotting allows zeros to be shown.

6.4 Limitations of the dataset

- **Legitimate websites were selected in a specific way:** all of them are homepages in the form https://www.domain with at least 102 lines of code. Real legitimate websites also include inner pages, HTTP websites and small pages.
- **Some features are partly circular:** URLSimilarityIndex is exactly 100 for every legitimate website, and IsHTTPS is 1 for all of them, which makes these features look stronger than they would be on new data.
- **Feature extraction errors:** in 79% of rows the URL features were calculated without the last character of the URL, and IsHTTPS was calculated incorrectly for 493 URLs (fixed in cleaning).
- **Placeholders and artifacts:** titles stored as "0" or "#NAME?", and IP numbers and ports stored as TLDs.
- **A snapshot in time:** phishing techniques change quickly, so these patterns may not hold in the future.

6.5 Recommendations for future analysis

- Collect legitimate websites that are more realistic (inner pages, HTTP websites, small pages) and compare the results.
- Build a predictive model without the circular features (URLSimilarityIndex and IsHTTPS) and compare it with a model that uses all features. Punjabi (2026) used a similar low-leakage idea, keeping only URL features and dropping the page-content features.
- Split the data by domain when training a model, so the same website does not appear in both the training and test sets.
- Test the model on obfuscated URLs (for example, subdomain injection and character replacement), since performance can drop a lot under these attacks (Punjabi, 2026).
- Analyze the Title text, for example with keyword or text analysis, to find words that are common on phishing pages.

6.6 Conclusion

This project explored 235,509 websites to find out what makes phishing websites different. Phishing websites tend to have longer and more complex URLs, very small pages, risky TLDs and almost no trust signals, while HTTPS alone is not a reliable sign of safety. At the same time, some of the strongest differences come from how the legitimate websites were collected, so the results are most useful as warning signs to combine, not as single rules. The dashboard makes these patterns easy to explore with filters and interactive charts.

7. References

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